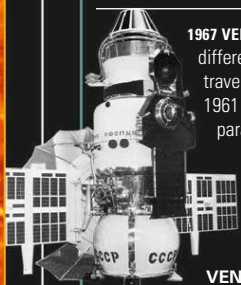


MISSIONS TO VENUS

Over 20 probes have investigated Venus. The first was the USA's Mariner 2, which made the first successful flyby of a planet in December 1962. Since then, probes have orbited Venus, plunged into its atmosphere, and landed on its scorching-hot surface.

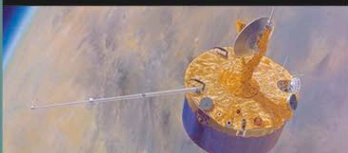


1967 VENERA 4 (USSR) Sixteen different Venera probes traveled to Venus between 1961 and 1983. Venera 4 parachuted through Venus's atmosphere in October 1967 and returned the information that it is primarily composed of carbon dioxide.

VENERA 4

1970 VENERA 7 (USSR) This was the first probe to make a controlled landing on the surface. An instrument capsule landed on the night side and measured the temperature.

1975 VENERA 9 AND 10 (USSR) The first image of the surface came from Venera 9. Its lander touched down on October 22, 1975, and returned an image of rocks and soil. Venera 10 did the same 3 days later.



1978 PIONEER VENUS (USA) Two Pioneer Venus probes, each with several components, arrived in 1978. An orbiter collected data that was used to make the first global map of Venus, and probes studied the atmosphere.

1981 VENERA 13 (USSR) Venera 13 survived on the surface for 2 hours 7 minutes on March 1, 1982. It took the first color images and analyzed a soil sample. Flat slabs of rock and soil can be seen beyond the edge of the probe in the image below.



1990 MAGELLAN (USA) Between September 1990 and October 1994, Magellan made four 243-day mapping cycles of Venus. It collected gravity data on the fourth cycle.

2005 VENUS EXPRESS (EUROPE) Launched in 2005, the European Space Agency's Venus Express went into a highly elliptical orbit (passing over the planet's poles) in 2006 to monitor its clouds, atmospheric circulation, and magnetic field. The mission lasted until fuel was exhausted in December 2014.



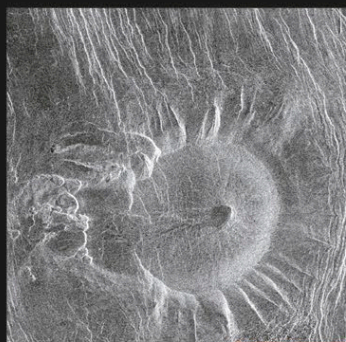
VENUS EXPRESS

2015 AKATSUKI (JAPAN) Achieved orbit around Venus 5 years late, after its initial orbit insertion burn ended prematurely when the engines unexpectedly cut out.

GEOLOGICAL FEATURES

Venus could be expected to have global features similar to those on Earth, but it differs in one key respect: it does not have moving plates. This means that its surface tends to move up or down rather than sideways. Yet Venus displays many familiar, Earth-like features formed by a range of tectonic processes, as well as some unfamiliar ones, such as arachnoids (see below). Venus has hundreds of volcanoes, from large, shallow-sloped shield volcanoes such as Maat Mons to small nameless domes. About 85 percent of the planetary surface is low-lying volcanic plains consisting of vast areas of flood lava. There has been volcanic activity as recently as about 500 million years ago, and it is possible that some of the volcanoes could be active. Other features are a result of the crust pulling apart or compressing.

There are troughs, rifts, and chasms, as well as mountain belts such as Maxwell Montes, ridges, and rugged highland regions. Venus's highest mountains and biggest volcanoes are comparable in size to the largest on Earth, but overall this planet has less variation in height.

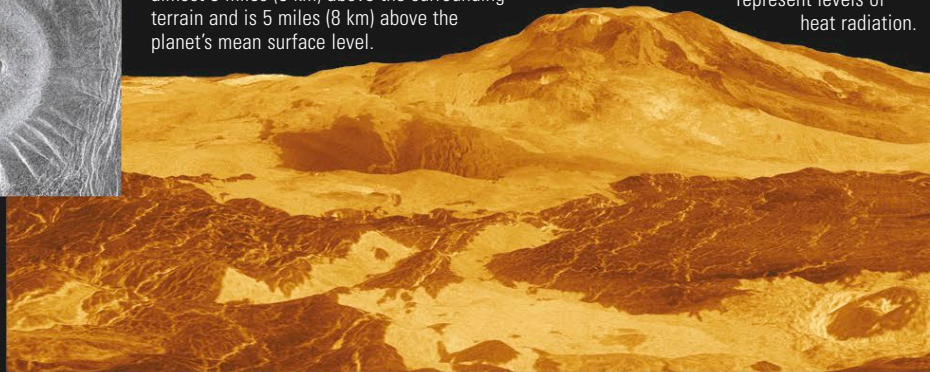


ARACHNOID

This spiderlike volcanic feature has a central circular depression (or dome) surrounded by a raised rim with radiating ridges and valleys.

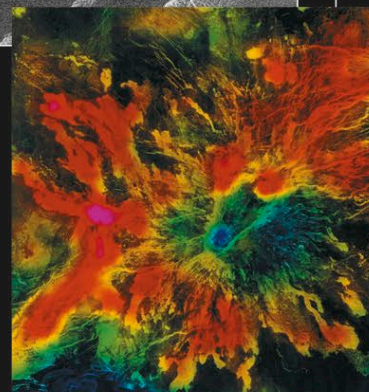
SHIELD VOLCANO

Venus's tallest volcano, Maat Mons, rises to almost 3 miles (5 km) above the surrounding terrain and is 5 miles (8 km) above the planet's mean surface level.



FRACTURES

This complex network of narrow fractures extends over about 30 miles (50 km) of northwest Aphrodite Terra. It is reminiscent of a river system on Earth, but the angular intersections indicate this is a tectonically formed system of fractures.



LAVA FLOWS

Solidified lava flows spread out for hundreds of miles in all directions from one of Venus's many volcanoes. The colors represent levels of heat radiation.

VENUS MAPS

These four views combine to show the complete surface of Venus. They have been labeled to show surface features, such as mountains, craters, highland regions, upland areas, and lowland plains.

